

FUAD AHMED (TUBA, FUAD AHMED)

STATEMENT OF PURPOSE & KAUST FELLOWSHIP MOTIVATION

Institution: King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia

Division: Computer, Electrical and Mathematical Sciences and Engineering (CEMSE)

Degree: Master of Science (MS) in Computer Science • **Applicant:** fuadahmedt@gmail.com • +251 925 278 350

To the Graduate Admissions Committee and Faculty of CEMSE at KAUST,

I am writing to formally present my application for admission to the **Master of Science in Computer Science** at **King Abdullah University of Science and Technology (KAUST)** and consideration for the prestigious **KAUST Fellowship** for the 2027/2028 academic year. As a software systems engineer with four years of hands-on industry experience building scalable distributed architectures and high-throughput data microservices, my objective is to conduct advanced research at the intersection of **Distributed Systems, High-Performance Computing (HPC), and Extreme-Scale Cloud Infrastructure**. KAUST's world-renowned computational facilities, culminating in the Shaheen III supercomputer and the Extreme Computing Research Center (ECRC), represent the pinnacle of global scientific computing where I aim to dedicate my graduate research.

1. BACKGROUND & COMPUTATIONAL FOUNDATIONS

My passion for computing systems originated from a fundamental fascination with concurrency and resource orchestration. I completed my Bachelor of Science in Computer Science at St. Mary's University in Addis Ababa (GPA: 3.20/4.00), within an intensive four-year curriculum conducted entirely in English. My undergraduate coursework grounded me in rigorous discrete mathematics, linear algebra, operating systems architecture, computer architecture, and networking protocols.

For my senior capstone project, I conceptualized and evaluated a high-concurrency, fault-tolerant distributed data synchronization protocol. Operating across networked Linux nodes, I implemented delta-encoded state serialization and lock-free memory ring buffers in Go and Python, ensuring state consistency across disparate nodes while drastically diminishing network payload overhead and CPU thread contention. This project cemented my conviction to dedicate my career to large-scale systems research.

2. PROFESSIONAL ENGINEERING EXPERIENCE & SYSTEMS RESEARCH DRIVERS

Over the past four years, I have served as a Lead Backend and Distributed Systems Software Engineer. I designed asynchronous microservice pipelines using Python (FastAPI/AsyncIO), Go, and Docker, serving concurrent enterprise workloads. In production environments, I architected distributed Redis caching topologies that reduced database read latency by 48% and engineered resilient message-driven pipelines utilizing Kafka and RabbitMQ to absorb volatile traffic spikes.

Working in emerging digital ecosystems where bandwidth and compute footprints are severely constrained taught me a vital lesson: computational scalability cannot rely on brute-force hardware provisioning. Instead, next-generation computing demands algorithmic efficiency, adaptive scheduling, and hardware-software co-design. To pioneer these breakthroughs, I must advance beyond applied software engineering into deep academic and experimental systems research.

3. WHY KAUST & CEMSE REPRESENT MY IDEAL RESEARCH CRUCIBLE

KAUST has established an extraordinary global reputation for fearless computational inquiry and international excellence. The CEMSE division provides an unparalleled ecosystem that is impossible to replicate elsewhere:

- **World-Class Supercomputing Infrastructure:** The presence of *Shaheen III* (HPE Cray EX system with cutting-edge GPU and CPU nodes) provides an unprecedented platform to test and evaluate extreme-scale distributed

algorithms, parallel communication frameworks, and memory-bound workloads under genuine petascale conditions.

- **Extreme Computing Research Center (ECRC):** I am deeply eager to align my research with the ECRC, specifically investigating scalable communication runtimes, distributed fault tolerance at extreme scale, and low-overhead consensus protocols that bridge cloud data centers with heterogeneous accelerator fabrics.

- **Interdisciplinary Collaborative Culture:** KAUST's international faculty and dedicated fellowship support create an environment completely unburdened by financial constraints, enabling graduate researchers to focus 100% of their intellectual energy on solving monumental computing challenges.

4. CAREER VISION & CONTRIBUTION TO KAUST

Upon earning my Master of Science degree at KAUST, I intend to continue toward doctoral studies or lead foundational research in high-performance cloud and distributed infrastructure within Saudi Arabia's burgeoning technological centers and global research institutions. Long term, I aspire to contribute to Saudi Arabia's Vision 2030 digital transformations and bridge computational innovation with developing economies in the Middle East and Africa.

I bring to KAUST exceptional technical tenacity, real-world systems architecture experience, and a deep appreciation for the university's international scientific mission. Supported by the KAUST Fellowship, I am prepared to contribute with distinction to the CEMSE research community.

Respectfully submitted,

Fuad Ahmed (TUBA, FUAD AHMED)

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